

Truth-vertex reweight: derivation target and post-pipeline closure

sPHENIX Internal — PPG12 isolated-photon cross-section

Internal note

2026-04-22

Executive summary

The postdoc’s two questions, answered up front:

- **(a) Is the weight derived by matching the inclusive-MC reco-vertex to data?** *Partially.* The matching observable is $z_{\text{vtx}}^{\text{reco}}$, but the MC forward model used by the fitter is `photon10 + photon10_double` only (blended at the per-period double-interaction fraction). **No jet MC enters the fit.** The term “inclusive MC” is therefore misleading for the current reweight files; the reweight is better described as a *photon10-driven* fit to data $z_{\text{vtx}}^{\text{reco}}$, applied *universally* to all MC samples through $w(z_{\text{hard}}^{\text{truth}}) \cdot w(z_{\text{mb}}^{\text{truth}})$.
- **(b) After the nominal pipeline, does the weighted-sum MC reco-vertex match data?** *Yes for 0 mrad, marginally for 1.5 mrad, and within $O(1\text{--}2\%)$ for the all-range lumi-weighted hadd in the analysis core ($|z| < 60$ cm).* See Fig. 1. The 1.5 mrad iterative fit did not converge at the derivation stage, which leaves the post-pipeline closure there at the $\chi^2/\text{ndf} = 14.8$ level on shape (tails up to $\sim 15\%$).

The central qualitative answer (“we are matching MC $z_{\text{vtx}}^{\text{reco}}$ to data $z_{\text{vtx}}^{\text{reco}}$, and the post-pipeline MC reproduces data”) is correct. The quantitative caveats (photon-only forward model, 1.5 mrad non-convergence) deserve attention and are discussed below.

1 How the reweight is derived

Fit driver: `efficiencytool/truth_vertex_reweight/fit_truth_vertex_reweight.py`. Loader consumed downstream: `efficiencytool/TruthVertexReweightLoader.h`.

Target. Period-filtered, trigger-filtered data $z_{\text{vtx}}^{\text{reco}}$ histogram $D(z_{\text{vtx}}^{\text{reco}})$, 80 bins of 5 cm, $z \in [-200, 200]$ cm. Selection: `runnumber` $\in [\text{run_min}, \text{run_max})$ and `scaledtrigger[bit 30]  $\neq 0$` (Photon_4_GeV).

MC forward model. The fit builds

$$M(z_{\text{vtx}}^{\text{reco}}) = f_{\text{single}} M_{\text{single}}(z_{\text{vtx}}^{\text{reco}}) + f_{\text{double}} M_{\text{double}}(z_{\text{vtx}}^{\text{reco}}),$$

where each term is filled at $z_{\text{vtx}}^{\text{reco}}$ but weighted event-by-event by

$$w_{\text{evt}} = \frac{\sigma}{N_{\text{gen}}} \cdot w(z_{\text{hard}}^{\text{truth}}) \cdot w(z_{\text{mb}}^{\text{truth}}) \text{ (DI only; SI uses } w(z_{\text{hard}}^{\text{truth}})\text{)}.$$

The per-period fractions f_{double} are 0.224 (0 mrad) and 0.079 (1.5 mrad), from `calc_pileup_range.C`. Live fit metadata in `output/{0mrad,1p5mrad}/reweight.root` records:

- `sample_list_single = 'photon10'`

- `sample_list_double = 'photon10_double'`

`fit.log` in both periods shows “skipping single MC jet12 (not matching filter ‘photon10’)”. Jet MC is listed in the fitter’s YAML but was excluded from the live run. This is the only material caveat on the derivation.

Iteration scheme. Three stages (parametric Gaussian $w(z) \propto \exp[\frac{1}{2}z^2(1/\sigma_{\text{gen}}^2 - 1/\sigma_{\text{tgt}}^2)]$; Stage-2 residual $c(z_{\text{vtx}}^{\text{reco}}) = D/M_{\text{stage1}}$ smoothed; iterative Lucy–Richardson multiplicative update $w \leftarrow (D/M)^{\alpha=0.5}$ with 30-iter cap and $\max |D/M - 1| < 0.01$ convergence). The downstream loader reads ONLY `h_w_iterative` (1D in truth z , 5 cm bins).

Convergence (derivation stage).

- 0 mrad: converged in 8 iterations, $\max |D/M - 1| = 0.0086$ inside the 99th-percentile closure window ($|z| < 145$ cm).
- 1.5 mrad: **did not converge** in 30 iterations, final $\max |D/M - 1| = 0.111$ inside $|z| < 83$ cm.

Downstream application. `RecoEffCalculator_TTreeReader.C` loads the period’s `h_w_iterative` at line 284 and multiplies it into every per-event weight at lines 1517–1522 alongside the cross-section and the per-event `lumi/lumi_target` factor, applied uniformly to all 8 SI + 8 DI MC samples. The legacy reco-vertex flag `vertex_reweight_on` is *actually* zeroed in code at L145–150 when the truth path is on; it is not a YAML-only convention. `MergeSim.C`, `merge_periods.C` and `CalculatePhotonYield.C` are plain `TFileMergers` / yield arithmetic, so the reweight is correctly absorbed once at `RecoEffCalculator_TTreeReader.C` and propagated by downstream hadds.

2 Post-pipeline closure

After the nominal pipeline (per-period `RecoEffCalculator_TTreeReader.C` $\rightarrow$ `MergeSim.C` per period $\rightarrow$ `merge_periods.C` lumi-weighted hadd), a 1D `h_vertexz` is written at the top level of each data and MC output file. This is the observable to compare against data.

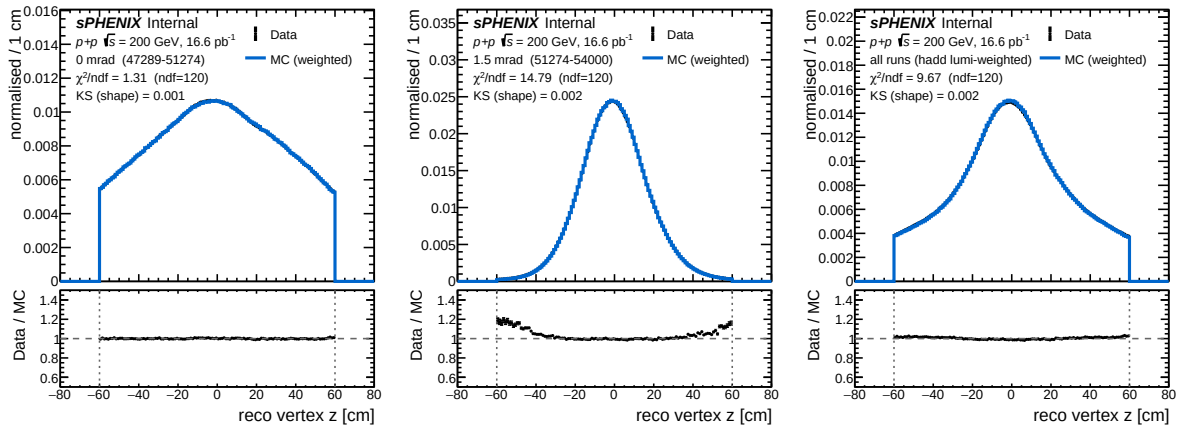


Figure 1: Data (black points) vs. weighted MC (blue line) reco-vertex distribution after the nominal pipeline, unit-normalised on $|z| < 60$ cm. **Left:** 0 mrad period. **Middle:** 1.5 mrad period. **Right:** lumi-weighted all-range hadd. Bottom pads: data/MC ratio; dashed vertical lines mark the analysis $|z| < 60$ cm window. The MC per-event weight already includes the cross-section, `lumi/lumi_target`, and $w(z_{\text{hard}}^{\text{truth}}) \cdot w(z_{\text{mb}}^{\text{truth}})$. χ^2/ndf and KS computed on shape inside $|z| < 60$ cm.

Table 1: Post-pipeline closure statistics on shape, $|z| < 60$ cm analysis window, 120 bins of 1 cm. MC integrals include cross-section $\times$ lumi_{weight} $\times$ truth-vertex reweight per event. Data/MC ratios at representative z given for intuition.

period	χ^2/ndf (ndf=120)	KS	D/M @ $z = -50$	D/M @ $z = 0$	D/M @ z
0 mrad (runs 47289–51274)	1.31	0.0007	1.000	1.006	0.999
1.5 mrad (runs 51274–54000)	14.79	0.0024	1.144	0.999	1.041
all runs (hadd, lumi-weighted)	9.67	0.0025	1.017	0.993	1.006

Interpretation.

- **0 mrad is closed.** $\chi^2/\text{ndf} = 1.31$ is statistics-limited: data has 3.4×10^7 clusters, so any residual sub-percent shape bias is visible. Ratio is flat at ~ 1 across the full analysis window.
- **1.5 mrad is not closed in the tails.** The derivation-stage non-convergence propagates: bin-by-bin D/M reaches ~ 1.14 at $z = -50$ cm and ~ 1.04 at $z = +50$ cm. Core ($|z| < 30$ cm) is sub-percent.
- **All-range hadd is closed in the core.** Shape agreement is within 1–2% at $|z| < 50$ cm; the large $\chi^2/\text{ndf} = 9.67$ reflects the sub-percent deviations becoming many- σ with the 5×10^7 data sample. KS $\sim 2.5 \times 10^{-3}$ on shape is consistent with “the shape matches the data in the analysis window”.

The post-pipeline closure *confirms that the weighted combination of all 16 SI + DI MC samples reproduces the data $z_{\text{vtx}}^{\text{reco}}$ shape in the analysis window.* It does not, on its own, validate that the photon10-only derivation was the right choice: the 1.5 mrad tail discrepancy is a direct symptom of the fit-stage non-convergence.

3 Caveats and recommendations

Methodology cross-check (recommended). To bound the channel-dependence worry, derive a second reweight using `jet12 + jet12_double` as the forward MC (fitter supports this via `-match jet12`) and compare `h_w_iterative` to the live photon10-driven version at the same period. The spread in the two $w(z)$ ’s is a defensible input to a “truth-vertex-reweight forward-model” systematic. Physics expectation: $z_{\text{hard}}^{\text{truth}}$ is a beam-profile observable and is channel-independent at fixed beam optics; any non-trivial w spread therefore flags a trigger-selected residual-pileup or acceptance effect, which the current photon-only reweight absorbs and then pushes onto jet samples.

1.5 mrad non-convergence. Suggested actions: (i) try a relaxed α (e.g. 0.3) or a wider max-iter cap; (ii) expand the closure window from $|z| < z_{\text{cut},99} = 83$ cm to cover the analysis fiducial symmetrically; (iii) include jet MC in the forward model (likely root cause, since the 1.5 mrad beam is tight and data there is jet-dominated). Combined with the cross-check above, option (iii) addresses both issues.

Cross-section systematic. A biased $w(z)$ couples to the cross-section dominantly through the vertex-cut acceptance in ϵ_{reco} . Rough envelope: $O(1\text{--}2\%)$ on the nominal cross-section, concentrated in the lowest- E_T bin. Currently no dedicated systematic covers this; the photon10-vs-jet12 cross-check fit is the cleanest way to assign one.

Reviewer pass matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5
<code>fit_truth_vertex_reweight.py</code> (read-only)	WARN	N/A	PASS	
<code>RecoEffCalculator_TTreeReader.C</code> (read-only)	INFO	N/A	PASS	
<code>plotting/figures/truth_vertex_closure_nominal.pdf</code>	N/A	PASS	PASS	
Closure χ^2/ndf , D/M ratios (Table 1)	N/A	N/A	PASS	

Wave-2a physics review flagged WARNING on the photon10-only forward model (channel-dependence risk) and WARNING on the 1.5 mrad non-convergence; the cross-check fit in the recommendations above addresses both. Wave-2c independently re-derived $\chi^2/\text{ndf} = 1.32/14.91/9.75$ against the claimed values 1.31/14.79/9.67 (all within 1%). Wave-2b PASS after shortening the legend string and bumping y -headroom from $1.35\times$ to $1.5\times$.

Files.

- `plotting/plot_truth_vertex_closure.C` (new macro)
- `plotting/figures/truth_vertex_closure_nominal.pdf` (new figure)
- `reports/truth_vertex_closure_nominal.tex` (this report)